A new modular open-hardware data-logger optimized for bicycle dynamics research

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Introduction

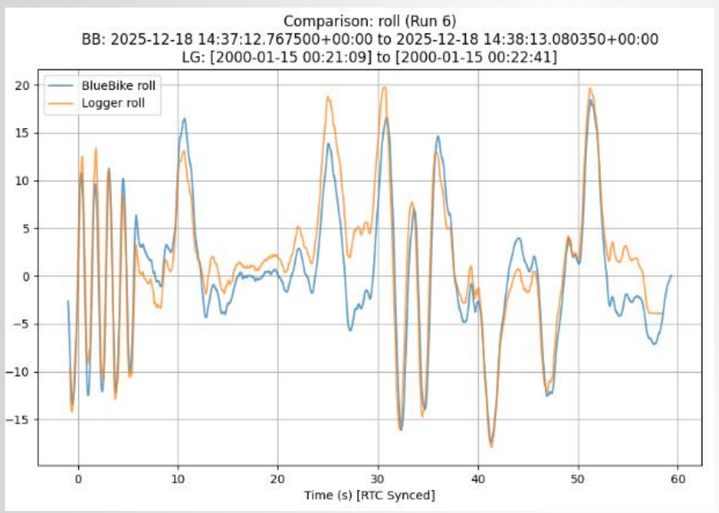
- More **bicycle** use → More questions → More research ≠ More **tools**
- High **resolution** and high **accuracy** measurements of bicycle **motion**
- Bicycle **crash research** requires robust and **reliable** equipment
  - **Inexpensive** in case something breaks

Challenges

- Difficult to reproduce custom-built systems
- Time-consuming development and maintenance
- Limited adaptability across experiments
- Barriers to sharing and reusing research tools

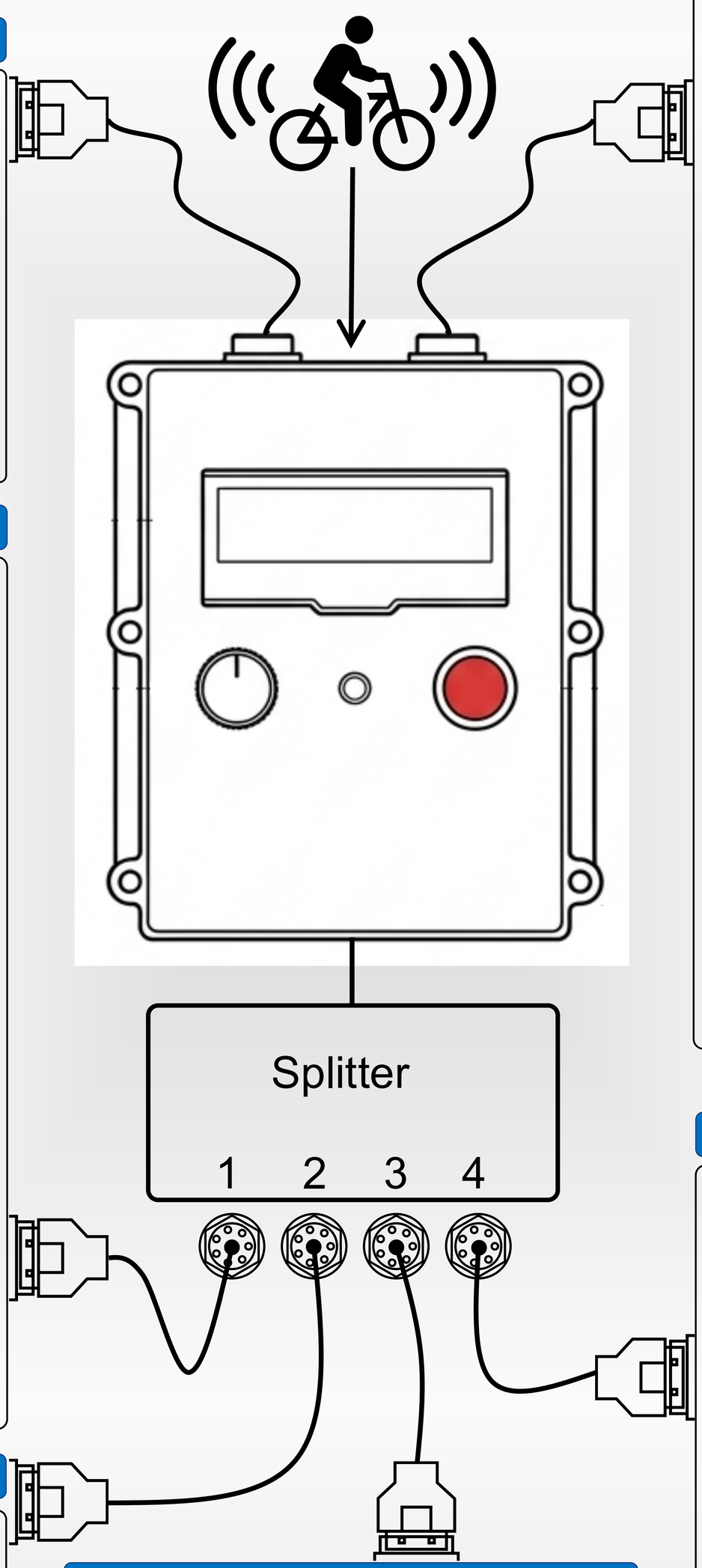
Impact & Results

- Reduced need to start from scratch
- More adaptable experimental setups
- Foundation for collaborative development
- Potential to improve reproducibility across studies
- Analog and digital sensors supported



What's next?

- Develop ready-to-use sensor modules
- Expand the range of supported measurements
- Improve setup and user documentation
- Simplify deployment and maintenance
- Encourage community adoption and contributions



Tools

- **GitHub** — version control and collaboration
- **Teensy** — signal processing
- **CAD** — hardware design
- **Python** — data analysis

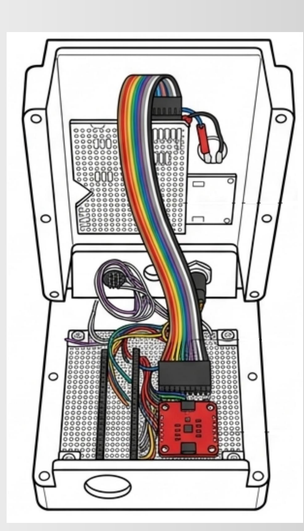
Acknowledgements

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Open by design

- **Open source:** hardware and software shared under the MIT license
- **Transparent:** publicly available design files and an open development process
- **Documented:** comprehensive technical documentation, setup instructions, and troubleshooting guides
- **Modular:** adaptable to different experimental requirements
- **Accessible:** built from off-the-shelf components
- **Maintainable**: easily replaceable components simplify repairs and upgrades
- **Collaborative:** community contributions and feedback are encouraged





Contribute & Connect

Take a look at the repo and **join us!**

github.com/mechmotum/open-data-logger

- Provide feedback
- Request modules
- Contribute to the documentation
- Suggest new ideas

